

In [3]:



```
1 import sqlite3
2
3 # Create (or reset) the database
4 conn = sqlite3.connect('sales_data.db')
5 cursor = conn.cursor()
6
7 # Drop the table if exists (clean start)
8 cursor.execute('DROP TABLE IF EXISTS sales')
9
10 # Create sales table
11 cursor.execute('''
12     CREATE TABLE IF NOT EXISTS sales (
13         id INTEGER PRIMARY KEY,
14         product TEXT,
15         quantity INTEGER,
16         price REAL
17     )
18 ''')
19
20 # Grocery sale Data
21 sample_data = [
22     # Rice transactions
23     ('Rice', 100, 40.0),
24     ('Rice', 120, 42.0),
25     ('Rice', 90, 41.5),
26     ('Rice', 150, 43.0),
27
28     # Wheat transactions
29     ('Wheat', 80, 30.0),
30     ('Wheat', 90, 32.0),
31     ('Wheat', 75, 31.0),
32     ('Wheat', 100, 33.5),
33
34     # Sugar transactions
35     ('Sugar', 60, 35.0),
36     ('Sugar', 70, 36.0),
37     ('Sugar', 50, 34.5),
38     ('Sugar', 80, 37.0),
39
40     # Tea transactions
41     ('Tea', 40, 150.0),
42     ('Tea', 50, 155.0),
43     ('Tea', 30, 152.0),
```

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44     ('Tea', 45, 158.0),
45
46     # Coffee transactions
47     ('Coffee', 30, 200.0),
48     ('Coffee', 25, 210.0),
49     ('Coffee', 35, 205.0),
50     ('Coffee', 40, 215.0),
51
52     # New Products
53     ('Dal', 60, 90.0),
54     ('Dal', 80, 95.0),
55     ('Oil', 20, 120.0),
56     ('Oil', 25, 125.0),
57     ('Salt', 100, 15.0),
58     ('Salt', 120, 16.0),
59     ('Spices', 15, 250.0),
60     ('Spices', 20, 270.0),
61 ]
62
63 cursor.executemany('INSERT INTO sales (product, quantity, price) VALUES (?, ?, ?)', sample_data)
64
65 # Commit and close
66 conn.commit()
67 conn.close()
```

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In [6]: ▶ 1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import sqlite3
4
5 # Connect to DB
6 conn = sqlite3.connect('sales_data.db')
7
8 # Query
9 query = '''
10     SELECT
11         product,
12         SUM(quantity) AS total_quantity,
13         SUM(quantity * price) AS total_revenue
14     FROM sales
15     GROUP BY product
16 '''
17 df = pd.read_sql_query(query, conn)
18
19 conn.close()
20
21 # Print results
22 print(df)
23
24 # Bar Chart
25 plt.figure(figsize=(10,6))
26 plt.bar(df['product'], df['total_revenue'], color='skyblue')
27 plt.xlabel('Product')
28 plt.ylabel('Total Revenue (INR)')
29 plt.title('Total Revenue per Product in India (Expanded Data)')
30 plt.xticks(rotation=45)
31 plt.tight_layout()
32 plt.show()
```

	product	total_quantity	total_revenue
0	Coffee	130	27025.0
1	Dal	140	13000.0
2	Oil	45	5525.0
3	Rice	460	19225.0
4	Salt	220	3420.0
5	Spices	35	9150.0
6	Sugar	260	9305.0
7	Tea	165	25420.0
8	Wheat	345	10955.0

